Loop Once, Think Twice: Adaptive Test-Time Compute for Molecular Structure Elucidation

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Abstract

Accurately inferring molecular structure from multimodal spectra remains difficult, and autoregressive SMILES decoding is especially brittle: a single wrong token invalidates the molecule. We tackle this with a compact Transformer that ingests IR,^1H-NMR and ^{13}C-NMR spectra in a single fused sequence. We also introduce an inner-layer "loop" during training, equiping the model with latent test-time compute, triggering extra forward passes using a fixed set of parameters. We also propose an acceleration-aware thresholding mechanism that allows the model to efficiently allocate latent compute during its forward passes. On a 39 M-parameter backbone, adaptive decoding lifts exact-match accuracy by 12 pp compared to its greedy decoding counterpart, at comparable wall-clock cost. The approach is model-agnostic and transfers compute-aware decoding techniques from natural language to fragile chemical languages.

1 Introduction

Photonics and spectroscopy have led to substantial progress in chemometrics, offering minimally invasive, cost-effective, and rapid alternatives to traditional chemical characterization methods [22]. Despite these advantages, techniques such as Near Infrared (NIR) Spectroscopy, Fourier Transform Infrared (FTIR) Spectroscopy, Nuclear Magnetic Resonance (NMR), and Raman Spectroscopy are still not comprehensively integrated into large-scale characterization pipelines. One reason is the continued reliance on human expertise for spectral preprocessing and interpretation, which often remains subjective and labor-intensive.

In recent years, there has been growing interest in machine learning (ML) and deep learning (DL) methods tailored to photonics and spectroscopy. These techniques have generally focused on classification and regression tasks (e.g., differentiating or quantifying molecular species) from raw or preprocessed spectra. However, the contemporary large-training paradigm associated with modern Transformer-based architectures—predominantly explored in Natural Language Processing—has not been systematically leveraged for spectral representation learning. The recent availability of large-scale, multimodal spectral datasets [3] now provides a novel opportunity to explore Transformer models for structure elucidation.

Motivated by these developments, we introduce a hybrid encoder-decoder and decoder-only Transformer framework that aims to autoregressively predict chemical structures (encoded as SMILES) from spectral data. In our approach, we utilize a ConvNeXt-based one-dimensional feature extractor, comparing its performance to more conventional one-dimensional convolutional networks in order to assess the impact of architectural choices.

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Preprint. Under review.

Beyond the model architecture, we show that pretraining on large-scale spectral data benefits considerably from employing the Muon optimizer over the widely used AdamW. Since Autoregressive SMILES prediction is fragile when compared to regular language modeling (meaning that sampling the wrong token has significantly worse outcomes, since the model has no way to backtrack) we explore the possibility of pretraining with looped layers [13] to induce latent test-time compute, as opposed to explicit test-time compute utilized by reasoning LLMs [6, 23].

2 Related Work

Deep learning has substantially advanced the inverse problem of molecular structure elucidation from spectral data. Traditional approaches often relied on rule-based systems or spectral databases, whereas recent methods use deep neural networks to learn direct mappings from spectra to molecular structures.

NMR and IR Spectroscopy. Transformer-based architectures have emerged as effective tools for predicting molecular structures from one-dimensional NMR spectra. Alberts et al. [2] proposed an encoder-decoder Transformer that predicts SMILES strings from ^{1}H and ^{13}C NMR spectra, achieving strong top-1 accuracy. The same team extended this approach to IR spectra, showing that full IR absorption profiles can be directly translated into molecular structures using similar Transformer-based architectures [1]. Wu et al. [28] proposed a patch-based self-attention mechanism to better capture local and global features in IR spectra. Hu et al. [16] introduced a multitask CNN-Transformer hybrid model that predicts both molecular formula and structure from NMR, highlighting the advantage of integrating multiple predictive tasks.

Autoregressive and Pretrained Models. Several works have explored autoregressive models, such as decoder-only Transformers fine-tuned on spectral data [15]. These models, inspired by chemical language modeling, benefit from pretraining on large molecular corpora and have been adapted to generate structures consistent with spectral constraints. Beam search, nucleus sampling, and entropy-aware decoding have all been explored to increase generation diversity and accuracy.

Graph-Based Methods and Diffusion Models. Beyond sequence generation, graph-based approaches such as DeepSPInN [8] model structure prediction as a reinforcement learning problem, where an agent incrementally builds a molecular graph conditioned on IR and NMR spectra. Recent advances also include DiffMS [4], which combines a Transformer encoder with a diffusion-based graph decoder conditioned on MS data, enabling the generation of chemically valid molecular graphs directly from spectra.

Multimodal Models. Recent efforts have combined multiple spectral modalities for improved prediction. Priessner et al. [24] proposed the MultiModalTransformer, which integrates 1D NMR, 2D NMR, IR, and MS data into a unified Transformer framework. Their results show that multimodal integration significantly improves accuracy, especially on real experimental spectra. Guo et al. [15] introduced the MolPuzzle benchmark to test multimodal reasoning, highlighting the limitations of general-purpose LLMs and the importance of domain-specific architectures.

Latent test-time compute and layer looping in Transformers Recent developments in RL-post-training of Large Language models, such as [23] and [6], have popularized the term "test-time compute" to indicate expenditure of Floating Point Operations to increase performance after training. This usually takes the form of either mechanically [12] performing branching and search in tokens space, or doing so based on a learned output structure, such as the <thinking> mechanism seen in DeepSeek R1.

Recurrence and early exiting latent test-time compute Recent works have introduced the concept of performing test-time compute in latent space by looping layers or blocks [14, 7, 11]. Such techniques show significant gains in performance before tokens are sampled, allowing models to incerase performance on-demand without needing extra parameters. This is achieved by ensuring the looped blocks are recurrent in latent space. Realizing recurrence in inner layer looping in transformers requires some carefully chosen architectural elements, many of which are detailed in [14]. The authors specifically note how recurrence is characterized by tokens in latent space converging to attractors in latent space. In their work, the authors propose two distinct methods for early exiting: projecting the latent representations to the vocabulary and exiting based on KL divergence with respect to the previous token, and exiting based on norm of the difference vector with respect to the previous token. We propose a second method that we find to be more practical and stable than those two, based on second order differences between steps.

Optimisers While *Adam* and its decoupled-weight-decay variant *AdamW* remain the default choice for large Transformer training, their element-wise adaptivity ignores matrix geometry and adds memory overhead. A recent alternative addresses these issues: **Muon** [17] performs semi-orthogonal updates via a Newton–Schulz step [25]. In 4 we benchmark Muon against AdamW [21], showing that Muon delivers the best performance for our task.

3 Methodology

We model input spectra via a mixed decoder-only and encoder-decoder architecture. Specifically, building on the work carried out by [3], we employ a peak-centered, discretized tokenization scheme for C-NMR and H-NMR spectra, while we utilize a convolutional architecture to extract features from raw pixels for the Infrared ones. The motivation behind this choice is related to the sparsity of the input spaces: NMR spectra rely on much sparser positional information about their peaks, and hence can rely on a discretized representation of the spectrum. IR spectra, on the other hand, have a much more distributed signal, for which convolutional techniques help in capturing local patterns while compressing the dimension of the features.

Dataset We build upon the publicly released multimodal spectroscopic dataset of Alberts et al. [3], which comprises 794 403 unique molecules extracted from the USPTO reaction corpus. Each entry consists of:

- **Infrared (IR)** profiles (400–4000 cm^{-1} at 2cm^{-1} resolution), simulated via GAFF-based MD and dipole autocorrelation;
- **NMR** data: 1D ^{1}H, 1D ^{13}C (with HSQC peak annotations) in $CDCl_3$, generated and auto-annotated in MestreNova;
- **MS/MS** spectra (positive and negative ESI; 10,20,40eV) from CFM-ID, SCARF and ICEBERG, with m/z–intensity and fragment formula annotations.

Molecules span 5–35 heavy atoms and cover a broad range of functional groups (alkanes, arenes, ethers, halides, etc.), facilitating both single-modality and truly multimodal structure-elucidation studies

Preprocessing We tokenize NMR spectra as described in [2], rounding positions to the second digit and inserting both intensity and positional information for each peak in the spectrum. IR spectra are instead interpolated using cubic splines and resampled over a grid of 1000 points in the 400 to 4000 cm^{-1} range.

Problem framing We frame the structure elucidation task as an autoregressive token prediction conditioned on spectral data.

Let $x = \{x_1, x_2, \dots, x_T\}$ be the SMILES tokens, and let s represent tokens encoding the spectral information. We factorize

$$p_\theta(x \mid s) \;=\; \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}, s),$$

where $x_{<t} = \{x_1, \dots, x_{t-1}\}$ and s is the preprocessed NMR spectra as described above. In practice, we maximize the expected log-likelihood:

$$\max_{\theta} \;\; \mathbb{E}_{(x,s)} \Bigg[\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}, s) \Bigg].$$

we model p_θ using a neural network, with an architecture detailed in the next section.

Model Architecture Our architecture can be expressed as an encoder for IR representations, followed by a multimodal decoder-only autoregressive transformer (training with causal mask) conditioned on a non-causally-masked prompt (combined encoded IR and tokenized NMR spectra). It differs from traditional encoder-decoder models in that all layers used for decoding can also modify the vector representations of the prompt; which follows latest practices on multimodal models.

ConvNeXT-1D IR Encoder To process raw IR spectra, we employ a one-dimensional adaptation of the ConvNeXT architecture [20]. This model modernizes traditional 1D CNNs by incorporating architectural choices inspired by the design of Vision Transformers (ViT) [10], while retaining the locality bias of convolutions.

The encoder is structured into four hierarchical stages with progressively increasing channel dimensionality and spatial downsampling, following a $(3, 3, 9, 3)$ stage-depth configuration and channel dimensions of $(96, 192, 384, 768)$. Each stage downsamples the input via strided convolutions and applies a stack of ConvNeXT blocks. The main components of each block are:

- **Patchifying Stem:** a non-overlapping convolution with kernel size 4 and stride 4, which acts as a patch embedding layer.
- **Depthwise Convolution:** ResNeXT-style channel mixing is achieved through depthwise convolutions.
- **Inverted Bottleneck:** the channel dimension is expanded prior to spatial mixing and reduced afterward, improving expressivity and computational efficiency.
- **Layer Normalization:** applied in place of traditional BatchNorm, stabilizing training across diverse signal scales.

This design allows ConvNeXT-1D to capture local structures while leveraging the depth and expressivity necessary to process the dense, information-rich patterns of IR spectra. We find this architecture to outperform standard 1D CNNs, which are widely used in spectroscopy [18], especially in modeling the highly distributed nature of IR signal patterns.

Multimodal Autoregressive Transformer Decoder. We implement a custom-masked autoregressive Transformer to handle both spectral (NMR, IR) and SMILES tokens. Specifically, the SMILES tokens follow a fully causal (unidirectional) mask, whereas the spectral tokens attend to each other bidirectionally. In other words, the decoder applies causal masking only among the SMILES tokens, but allows full attention among the spectral tokens to capture their interdependence (see Figure 1).

In addition, we incorporate RoPE [26] to encode positional information for all tokens. Let the inputs be x_{NMR} and x_{IR}, corresponding to the NMR and IR spectra, respectively. First, the IR spectrum is passed into an encoder network:

$$y_{\mathrm{IR}} \;=\; f_\theta^{\mathrm{encoder}}\big(x_{\mathrm{IR}}\big) \;\;\in\; \mathbb{R}^{d\times \frac{\ell}{32}},$$

where d denotes the hidden dimension and $\frac{\ell}{32}$ the downsampled length dimension of the IR representations. We then form a "prompt" by concatenating x_{NMR} and y_{IR}:

$$p \;=\; x_{\mathrm{NMR}} \,\oplus\, y_{\mathrm{IR}},$$

and feed p into the autoregressive Transformer decoder. During SMILES generation, the spectral tokens $(x_{\mathrm{NMR}}, y_{\mathrm{IR}})$ can attend to one another fully, while the SMILES tokens attend only to themselves in a strictly causal manner. As suggested in [14], we use RMSnorm [29] as a normalization in a sandwhich norm scheme.

Latent Test-Time Compute Given our output spaces's fragility, we explore ways to reliably increase the quality of the model's prediction without sampling tokens. To do so, we implement the block-wise layer-looping as described in [14]. Specifically, we select the innermost block in the transformer model to be repeated a fixed number of times per forward pass. This number is sampled as $k \sim U([0, n])$ during model training, so that the model learns to perform recurrence across arbitrary amounts of loops to improve its internal representations at test time. This approach is instrumental to properly produce a pipeline that can reliably guides the model towards a more certain output distribution, without relying solely on adjusting temperature settings, but by actually improving the model's internal representation.

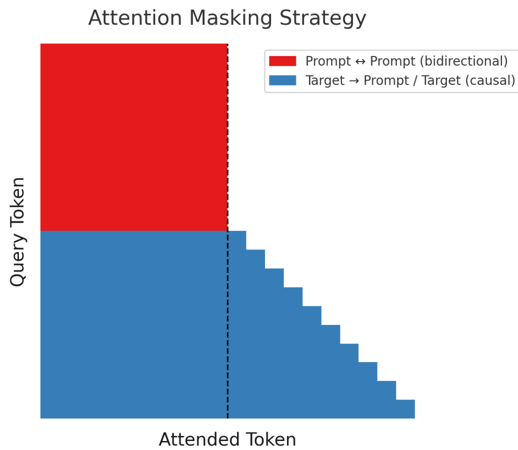


Figure 1: Attention mask scheme used in the decoder: spectral tokens use a bidirectional mask, while SMILES tokens follow a causal mask.

Block taxonomy We partition the Transformer into four conceptual modules:

1. *Embedding table* $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$ that maps input tokens to d-dim vectors.
2. *Encoding blocks* (L_{enc} layers)
3. *Hidden blocks* (L_{hid} layers) these lie "in the middle" of the stack.
4. *Decoding blocks* (L_{dec} layers) that specialise to causal dependencies among molecular tokens and terminate in a tied projection $\mathbf{E}^{\top}$.

Choosing the innermost block. Let $L = L_{\text{enc}} + L_{\text{hid}} + L_{\text{dec}}$ be the total number of Transformer layers and index layers from 1 (closest to the embeddings) to L (closest to the output head). We designate the *innermost* block

$$f_\theta \;=\; \text{TransformerLayer}_{\,\ell^\star}, \qquad \ell^\star = L_{\text{enc}} + \left\lceil \tfrac{L_{\text{hid}}}{2} \right\rceil.$$

Looping mechanism. During *training*, we sample a loop count $k \sim \mathcal{U}\{0,\, n\}$ (we use $n = 30$ unless stated otherwise). For a token sequence producing hidden state $\mathbf{h}_{\text{in}} \in \mathbb{R}^{T \times d}$, the looped dynamics are

$$\mathbf{h}^{(0)} = \mathbf{h}_{\text{in}} \tag{1}$$

$$\boldsymbol{\epsilon}^{(1)} \sim \mathcal{N}\big(\mathbf{0},\, \sigma^2 \mathbf{I}\big) \tag{2}$$

$$\mathbf{h}^{(1)} = f_\theta\big(A_\phi\big[\mathbf{h}^{(0)} \,\|\, \boldsymbol{\epsilon}^{(1)}\big]\big) \tag{3}$$

$$\mathbf{h}^{(t)} = f_\theta\big(A_\phi\big[\mathbf{h}^{(0)} \,\|\, \mathbf{h}^{(t-1)}\big]\big), \quad t = 2, \ldots, k \tag{4}$$

$$\mathbf{h}_{\text{out}} = \mathbf{h}^{(k)} \tag{5}$$

with A_ϕ being a RELU-activated projection.

Full hyper-parameters for looping (n, σ, scheduling) are reported in Appendix A.1.

Sandwich norm and latent recurrence As pointed out in [14], we find that regular pre-normalization of operations in the transformer blocks [30] is insufficient for a proper recurrence mechanism to be learned by the model. For this reason, we instead utilize the sandwich normalization scheme, similar to strategies found in [27, 9]

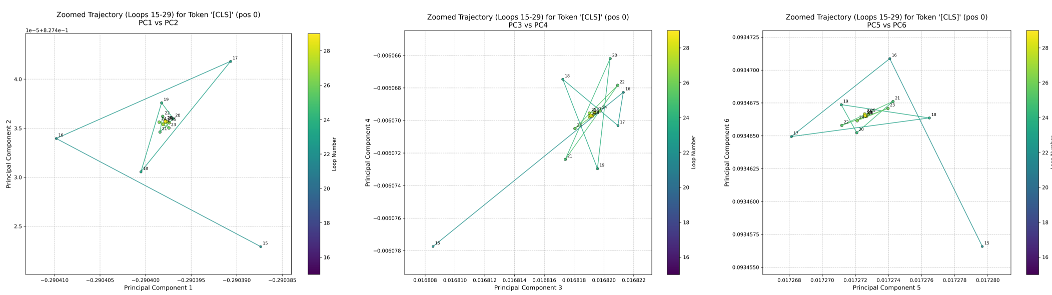
Adaptive test-time compute Geiping et al. [14] propose two strategies to exit early from block-looping. Let X_t represent activations of a block looped t times, and let $\mathbf{E}^T$ be the transformer's decoding head.

- KL divergence based, where we set ϵ_{KL} as a threshold, and exit if $KL(\mathbf{E^T}(\mathbf{X_t}), \mathbf{E^T}(\mathbf{X_t - 1})) < \epsilon_{KL}$
- Difference based, where the criterion for exit $\max(\|X_t - X_{t-1}\|_2) < \epsilon_{norm}$, given some threshold ϵ_{norm}.

While both techniques are intuitive and potentially effective, each presents drawbacks: the KL divergence exit mechanism requires effectively materializing the output distribution at every loop of the inner recurrence. This may become computationally expensive, especially in the case of large vocabulary sizes. While the second mechanism avoids projecting over the vocabulary, both from Geiping et al.'s work and our reproduction of the experiment for our setting, it can be seen that tokens don't always converge to a fixed point, and instead lie on orbits of varius sized. this may make a fixed threshold too brittle of a mechanism to be a reliable proxy for saturation of the latent recurrence-driven performance increase. We propose the following mechanism:

Acceleration-based exit exit if $max\|\delta_t - \delta_{t-1}\| < \epsilon_{acc}$ with $\delta_\tau = \|X_\tau - X_{\tau-1}\|$

This choice is intuitively justified by noticing we are estimating the second order time difference of convergence of the representation, which may be thought of as a local measure of their "acceleration" in latent space. This is meant to better detect conditions of regular periodic orbits, such as the one visualized in Fig. 2



(a) PC1 vs PC2 (b) PC3 vs PC4 (c) PC4 vs PC5

Figure 2: Pairwise projections of the last 15 recurrent steps of a single-token latent-space trajectory, following the approach in the appendix of [14]. (a) PC1–PC2, (b) PC3–PC4, (c) PC4–PC5.

Optimizers We choose to utilize the Muon optimizer [17], based around the idea of leveraging the matrix structure of weights to semi-orthogonalize matrix-shaped parameters' updates via a newton-schulz iteration. This can also be seen as a way of doing steepest descent under the Spectral norm, instead of the regular Frobenius one [5]. Such a decision, apart from recent literature on the topic [19], is suggested by our experiments 5. For non-matrix weights, we utilize AdamW [21]

4 Experiments

We focus on evaluating structure elucidation from the combination of IR and NMR spectra. First, we ablate a series of architectural choices over a given token horizon. Then, we test the resulting architecture at a lower or equal computational requirement with respect to the current state of the art on the dataset. We train a significantly larger model and set the state of the art for the task on this dataset. We then show performance increasing reliably with test-time compute in the shape of layer looping. We finally explore a set of classical decoding strategies and compare it to the adaptive latent test-time compute paradigm.

Modality and optimizer ablations We ablate the following configurations:

- Muon

- AdamW
- IR-only (Muon)
- NMR-only (Muon)

This allows us to demonstrate the importance of each modality in the structure elucidation task, as well as deciding on the better optimizer.

Experimental Setup We fix the model depth to 8 layers, with 4 heads for every attention operation, while the embedding dimension is set to 512. for the ablations, we sweep learning rates for optimizers on a fixed 50 thousand iterations budget, with a batch-size of 32. We train with a linear warmup of 1000 steps and a cosine decay schedule, decreasing to $\frac{\eta_{peak}}{100}$, where η_{peak} is the peak learning rate. We run two full training runs at 5 epochs of training with a 90-9 % split between train and test. We keep a 1% set for validation. We run all experiments with 0.1 weight decay and 0 dropout.

Test-time compute scaling We observe a consistent improvement in performance with increasing test-time compute induced via latent looping. Table 1 summarizes the exact match rates across different loop counts on the test set for the model, while 3 shows an example of our model predicting structure at different looping regimes.

Table 1: Metrics (rows) at various loop counts (columns) for different pretraining steps. Best values (furthest loop among ties) are **bolded**. Experiments are performed over 50 randomly sampled molecules from the test set.

	Num Loops					
Metric	1	5	15	30	45	60
			10 K steps			
Valid SMILES (%)	64.00	84.00	84.00	84.00	84.00	**84.00**
Exact Match (%)	8.00	20.00	20.00	20.00	20.00	**20.00**
Tanimoto	0.3394	**0.4836**	0.4829	0.4829	0.4829	0.4829
MCS Ratio	0.4647	0.6353	0.6366	0.6366	0.6366	**0.6366**
ECFP6 IoU	0.2876	**0.4186**	0.4168	0.4168	0.4168	0.4168
			50 K steps			
Valid SMILES (%)	72.00	92.00	96.00	96.00	96.00	**96.00**
Exact Match (%)	24.00	32.00	36.00	36.00	36.00	**36.00**
Tanimoto	0.4748	0.6157	0.6468	0.6468	0.6468	**0.6468**
MCS Ratio	0.5849	0.7449	0.7867	0.7867	0.7867	**0.7867**
ECFP6 IoU	0.4344	0.5541	0.5879	0.5879	0.5879	**0.5879**
			100 K steps			
Valid SMILES (%)	84.00	88.00	92.00	92.00	92.00	**92.00**
Exact Match (%)	32.00	**36.00**	32.00	32.00	32.00	32.00
Tanimoto	0.5281	0.6623	0.6649	0.6764	0.6764	**0.6764**
MCS Ratio	0.6672	0.7839	0.8095	0.8095	0.8095	**0.8095**
ECFP6 IoU	0.4812	0.6155	0.6089	0.6178	0.6178	**0.6178**

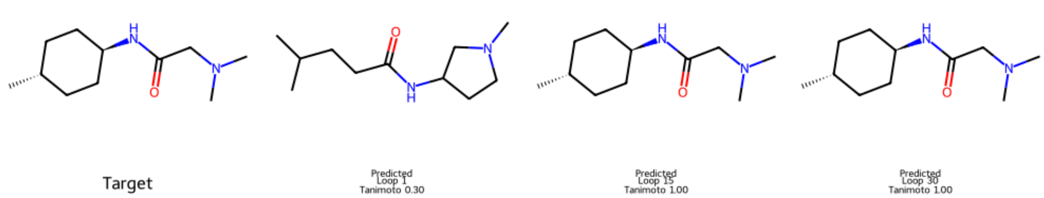


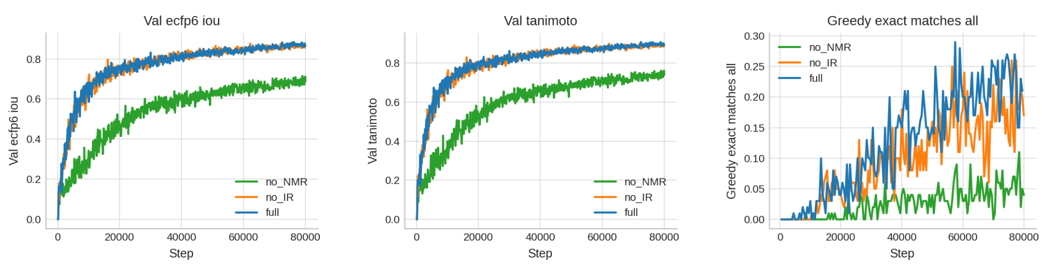
Figure 3: Examples of predictions by our model over different looping regimes.

Adaptive decoding In 2 we compare regular sampling methods with our adaptive latent test-time compute mechanism. On a comparable wall-clock budget, adaptive looping yields clear gains across all structural metrics.

Table 2: Comparison of decoding strategies on the validation set. STE = standard error. Arrows indicate desirable direction (↑ higher is better, ↓ lower is better). All results are based on 100 random samples from the test set. The model used is the 100k steps checkpoint discussed in this section. Greedy Loop represents greedy decoding with a maximum of 30 loops always utilized. As we can see, the adaptive compute method preserves most of the performance at a much lower inference time. We select $\epsilon_{acc} = 10^{-3}$.

Method	Valid (% ↑)	Match (% ↑)	Tan. (↑)	MCS (↑)	IoU (↑)	Time (s ↓)
Greedy	88.00±6.50	28.00±8.98	0.5588±0.0694	0.7157±0.0630	0.5011±0.0725	1.1075
Sampling	76.00±8.54	8.00±5.43	0.4733±0.0694	0.6290±0.0767	0.4157±0.0681	0.7221
Nucleus	76.00±8.54	24.00±8.54	0.5338±0.0793	0.6636±0.0796	0.4910±0.0795	0.7391
Adaptive	88.00±6.50	32.00±9.33	0.6557±0.0648	0.7781±0.0625	0.6019±0.0675	1.4729
Greedy loop	92.00±5.43	32.00±9.33	0.6764±0.0594	0.8095±0.0538	0.6178±0.0635	166.3930

Ablations As discussed in the previous section, we run ablations over multimodal use of the models. In Figure 4 we compare the baseline model to the `no-IR` and `no-NMR` ablations over a short (50 thousand steps) training horizon.



(a) `val_ecfp6_iou` (b) `val_tanimoto` (c) `exact match rate`

Figure 4: Ablation over modality usage: baseline (full) vs. no-IR and no-NMR variants. Metrics are smoothed via a moving average over 1k steps. Validation metrics are calculated over validly generated molecules only.

We also perform a sweep over a set of learning rates to validate our choice of optimizer. In figure 5, we can see how Muon is significantly more sample efficient than AdamW, as also demonstrated for NLP by [19].

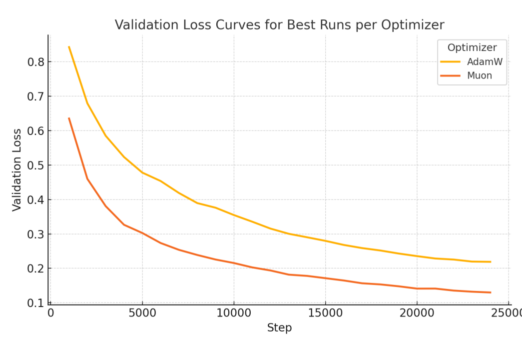


Figure 5: Best loss functions for Muon (10^{-3}) and AdamW (10^{-4}), across a sweep of different learning rates(in ther range $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ with AdamW lr in the non-Muon updated parameters always chosen of one tenth of the Muon one, over one epoch of the dataset at a batch size of 32

5 Discussion

Key take-aways. Our study demonstrates that *latent test-time compute*—implemented as a fixed-depth inner-layer loop—can substitute, in part, for additional parameters when molecular-scale Transformers are constrained by memory or deployment budgets. On a 39 M-parameter model, as few as 15 refinement loops lift exact-match by **12 pp** and performance saturates well before the maximum of 60 iterations. Because looping reuses weights, model size on disk and in memory is unchanged, and the mechanism integrates seamlessly with existing chemometric pipelines.

Limitations. *(i) Sim–to–real gap.* Our benchmark relies on high-fidelity *simulated* IR and NMR spectra; real instruments introduce peak asymmetry, solvent shifts and variable SNR. Bridging this gap will require fine-tuning on curated experimental spectra or domain-adaptation objectives that enforce spectral realism. *(ii) Output fragility.* SMILES is brittle—one token error invalidates an entire molecule. Looping refines hidden states but cannot correct a committed mistake; robust representations (e.g. SELFIES) or graph decoders remain attractive complements. *(iii) Variable latency.* While parameter-free, looping incurs input-dependent compute. High-throughput screening pipelines will need to trade off additional iterations against total throughput.

Broader impact. Affordable, reliable structure elucidation has the potential to broaden access to spectroscopic techniques, speed up reaction monitoring, and support automated synthesis workflows. At the same time, it's important to recognize that simplifying molecular prediction also calls for thoughtful oversight to ensure it isn't misused for harmful purposes

Future work. **Adaptive halting**—learned or norm-based—could trim superfluous loops without degrading accuracy. **Physics-aware decoding** that enforces valence or energy constraints would focus latent compute on chemically plausible candidates. Finally, scaling to richer modalities (2D-NMR, MS/MS, time-resolved spectra) will test whether looping continues to provide favourable accuracy–FLOPs trade-offs as the input bandwidth grows.

Code availability

Github with code will be released after double-blind review for anonimity reasons.

Acknowledgements

Computational Resources

Experiments were performed on NVIDIA T4 (14 GB VRAM) and RTX 4090 (24 GB VRAM) GPUs.